Integrated Environmental Assessment and Management, 2026, 1–10
https://doi.org/10.1093/inteam/vjag007
Published: January 10, 2026
Brief Communication | Special Series | Life Cycle & Sustainability

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Microfibers in life cycle assessment: comparing the physical effects of cellulosic and synthetic fibers via characterization factors development

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Editor's Note: This article is part of the special series "Effects of Textiles Microfibers and Chemicals in the Environment." Articles in this special series aim to answer the questions: How do we build a systematic understanding of the main fiber leakage points across different textile applications important to the global economy? What methods are needed to track fiber-based items and the microfibers they release through the waste stream and into the environment (air, soil, water)? What factors influence the ecotoxicology of microfibers (natural, anthropogenically modified, synthetic) and how can we study them? Can we design safer alternative materials, feedstocks, manufacturing, and chemical processes to minimize environmental degradation and preserve natural resources?

Abstract

Textile clothing is a significant contributor to microfiber pollution in aquatic ecosystems. A large portion of these emissions are cellulosic-based, either natural (e.g., cotton, linen) or semisynthetic (e.g., viscose, lyocell), which can be ingested by aquatic organisms, posing harmful effects even at environmentally relevant concentrations. Studies comparing the effects of cellulosic and synthetic fibers report conflicting results: some suggest synthetic fibers are more harmful, others the opposite. However, despite their high environmental abundance, the lack of a species sensitivity distribution (SSD) for cellulosic microfibers (CMFs) has prevented comparison of their effects with synthetic microfibers (SMFs). Further, although SMFs have been integrated into life cycle assessment (LCA) through characterization factors (CFs), equivalent CFs for CMFs are missing due to a lack of an exposure and effect factor (EEF) specific to these fibers. We derived CFs for six CMFs (cotton, linen, viscose, lyocell, rayon, and modal) by combining modeled fate factors in marine water and sediments with an EEF calculated from a hazardous concentration for 20% of species (HC20), using an SSD of EC10eq values. Taxonomically split SSDs were also obtained for further analysis of the mode of action of CMFs and microplastics. The HC20 for CMFs was not significantly different from that of microplastics, suggesting similar physical effect mechanisms. Fate modeling indicated lower persistence for CMFs, resulting in CFs one to two orders of magnitude lower than those for SMFs. Computed CFs were applied in an LCA comparing the ecosystem quality impacts of a cotton and a polyester tee shirt while accounting for microfiber emissions. Results show that polyester emissions caused significant impacts compared with other life cycle impacts, whereas cotton emissions did not. This work therefore provides the first comparative LCA of a cellulosic and a synthetic textile that considers impacts of microfiber emissions, and computes CFs compatible with different life cycle impact assessment methods.

Keywords environmental footprint, marine litter, toxicology, persistence, contaminants of emerging concern

Introduction

Textile clothing is estimated to cause the release of ~2 million tonnes of microfibers into the marine environment each year (Boucher & Friot, 2017). Microfibers are defined as fibers with a diameter < 5 mm (Scientists' Coalition for an Effective Plastics Treaty, 2025). A large portion of emitted microfibers are anthropogenically modified cellulosic-based, either natural (e.g., cotton and linen) or regenerated cellulose (also called semisynthetic fibers, e.g., viscose and lyocell). When emitted, these small particles are readily ingested by marine organisms in water and sediments, posing harmful effects even at environmentally relevant concentrations (Acharya et al., 2021; Siddiqui et al., 2023; Walkinshaw et al., 2023). Although it is expected that cellulosic microfibers (CMFs) will degrade much faster in the environment than microplastics (MPs) and synthetic microfibers (SMFs), little is known about their effects (Zambrano et al., 2020). Several studies comparing the effects of CMFs and synthetic fibers report conflicting findings: some suggest that cellulosic fibers are more harmful than synthetic (D. Kim et al., 2021), whereas others indicate the opposite (dos Santos et al., 2024; Siddiqui et al., 2023; Walkinshaw et al., 2023), and some report similar effects (L. Kim et al., 2021; Mateos-Cárdenas et al., 2021). Yet, these findings were not supported

Received: July 30, 2025. **Revised:** January 7, 2026. **Accepted:** January 8, 2026.
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by species sensitivity distribution (SSD) curves. These curves allow for comparison of the physical effects or ecotoxicity of different pollutants while encompassing the sensitivity of several species (Posthuma et al., 2002).

Life cycle assessment (LCA) is an environmental decision-making tool for quantifying emissions and resource extraction, associated with a product or service throughout its life cycle, along with their potential impacts on ecosystem quality and human health (as well as resource depletion/ecosystem services). In LCA, efforts of the MarILCA (MARine Impacts in LCA) group have led to the creation of characterization factors (CFs) for microfibers, which link the release of fibers to their damage on ecosystem quality (EQ) by combining their fate factors (time-integrated mass of pollutants in the environment) and physical effects (derived from an SSD curve; Corella-Puertas et al., 2023; Saadi et al., 2025). Most LCAs of textiles in the literature did not include the impacts of leaked microfibers, as life cycle impact assessment (LCIA) methods did not consider their impacts until recently (Watson & Wiedemann, 2019). Nonetheless, LCA studies of textiles that have included the impacts of leaked MPs have shown that emissions of SMFs over the life cycle of textiles can be a significant contributor to EQ damage (Pellengahr et al., 2025; Saadi et al., 2025). Further, LCAs of textiles that compared synthetic and natural materials indicate higher impacts on EQ of natural textiles but did not include the impacts of emitted CMFs or SMFs (Van Der Velden et al., 2014). This is particularly important, as it is suggested that natural and regenerated cellulosic textiles release more microfibers during washing and are sometimes found in higher abundance in the environment than SMFs (Acharya et al., 2021; Athey & Erdle, 2022; Stanton et al., 2019; Zambrano et al., 2019). Additionally, several studies found a greater abundance of anthropogenically modified cellulose fibers than natural cellulose or MPs in multiple marine species (Dampe Acharige et al., 2025; Traylor et al., 2024).

Only preliminary CFs were proposed for CMFs due to the lack of exposure and effect factor (EEF_{CMF}) specific to these fibers (Saadi et al., 2025). Current LCIA methods therefore overlook potential environmental impacts from CMFs, which are often believed to be more sustainable than synthetic fibers. It is therefore essential to develop CFs specific to CMFs to allow for more meaningful assessments and comparisons between products that emit synthetic or cellulosic microfibers during their life cycle, such as textiles, personal products, and cigarette butts (Athey et al., 2022; Weis & De Falco, 2022).

Thus, this work aims to (a) gather effect data from the literature to build the SSD curve for CMFs and compare it to that of MPs, (b) compute the $EEF_{CMF,}$ (c) combine the EEF_{CMF} with a fate model for microparticles to derive midpoint and endpoint CFs for CMFs for use in LCA, and (d) contextualize the potential impacts of CMFs and SMFs by applying the CFs in an LCA case study comparing a cellulosic and a synthetic tee shirt.

Methods

In LCIA, EQ CFs link an emission of a pollutant to its potential impacts on EQ and are usually computed by multiplying fate factors (FF) and EEF (Owsianiak et al., 2023; Rosenbaum et al., 2007).

Fate model overview

The model developed by Saadi et al. (2025) was used to compute FFs for CMFs in this work. The model considers water and sediments compartments at the continental scale. Fate mechanisms include degradation, sedimentation, deep burial, and resuspension and are quantified to obtain the time-integrated mass of CMFs in the marine environment (expressed in [$kg_{compartment}$ d/$kg_{emitted}$]; Rosenbaum et al., 2007; Saadi et al., 2025).

For the degradation rate, the model considers the surface-specific degradation rate (SSDR, expressed in [μm/d]) approach developed by Maga et al. (2022) and adapted by Corella-Puertas et al. (2023). Sufficient degradation rate data in marine water were found for cotton, linen, viscose, rayon, lyocell, and modal (S. Kim et al., 2022; Nagamine et al., 2022; Royer et al., 2023; Zambrano et al., 2019, 2020); however, data are lacking for sediments, where the SSDR was assumed equal to that in water. Although degradation in sediments is likely slower because of lower oxygen availability and reduced microbial activity, no consistent quantitative relationship is available to parameterize this difference across polymer types (Maga et al., 2022). As more degradation data become available, additional FFs and CFs can be computed for other CMFs. The study of Corella-Puertas et al. (2023) also provides density-dependent sedimentation rates, whereas the deep burial and resuspension rates are based on the model of Quik et al. (2023).

EEFs

Exposure and effect factors, expressed as a potentially affected fraction of species (PAF) integrated over volume (PAF $m^3/kg_{compartment}$), link the fate of a pollutant ($kg_{compartment}$ d/$kg_{emitted}$) in a compartment to an impact (PAF m^3 d/$kg_{emitted}$) and are developed from SSD curves (Fantke et al., 2017; Owsianiak et al., 2023). Saadi et al. (2025) used as a preliminary assumption that the EEF_{CMF} equals 50% that of MPs (EEF_{MP}), based on a study that found that cotton microfibers induced a decrease in growth in juvenile mussels that was ∼50% lower than that caused by the same concentration of polyester microfibers (Walkinshaw et al., 2023). As more data are now available, an EEF could be computed specifically for CMFs. The SSD curve is used to compare the physical effects of CMFs and SMFs, based on the sensitivity of multiple species (Posthuma et al., 2002).

A review of the effect concentration literature was done to build an SSD curve for cellulosic materials, based on effects on survival, growth, and reproduction (Owsianiak et al., 2023). Details of the keyword search can be found in online supplementary information supporting information A. Following updated recommendations, the lognormal fitted SSD curve (on a mass basis, [mg/L]) was derived from EC10 data (effect concentration 10%), and the HC20 (hazardous concentration for a 10% effect in 20% of the species) is used to compute EEF_{CMF} ($EEF_{CMF} = 0.2/HC20_{EC10}$; Owsianiak et al., 2023). When EC10 data for a species were not available in a study, lowest observed effect concentrations (LOECs) and no observed effect concentrations (NOECs) were converted to equivalent 10% effect concentrations (EC10eq) using extrapolation ratios of Aurisano et al. (2019). These extrapolations were originally derived for chemicals; however, we applied them to microfibers, as they rely on empirical correlations of effect thresholds rather than on a specific mode of action. Further, we followed the methodology of Lavoie et al. (2022) to compute the 95% confidence interval around the EEF_{CMF}. The calculations can be found in online supplementary information supporting information B.

Lavoie et al. (2022) reported no significant effect of particle size or shape of the SSD curve of microplastics. Due to limited effect data, a similar shape and size-specific analysis could not be conducted for

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cellulosic materials. As a result, only a single SSD curve encompassing all types of cellulosic materials was developed. Additionally, SSD curves split by taxonomic groups (vertebrates, invertebrates, and microalgae or aquatic plants) for MPs and CMFs were generated following the methodology of Oginah et al. (2023) for further analysis of the mode of action. In LCIA, it is common to generate a single SSD curve from all available effect data (Fantke et al., 2018; Owsianiak et al., 2023), but a taxonomically differentiated approach is relevant for risk assessment or interpretation of sensitivity differences among taxonomic groups for better environmental decision-making (Oginah et al., 2023).

Extrapolation of EEF

Sufficient data points were available in the literature only for aquatic (water-dwelling) species, yet the potential impacts in sediments were found to also be particularly relevant in the case of nonbuoyant particles, which settle quickly (Saadi et al., 2025). Thus, the EEF_{CMF} computed was extrapolated to the sediments using the ratio between microplastics EEFs in water ($EEF_{w,\ pol}$) and sediments ($EEF_{sed,\ pol}$) calculated by Saadi et al. (2025). The $EEF_{sed,\ pol}$ for MPs was found to be approximately two orders of magnitude smaller for sediments than for water, and it was shown that the difference was due to a difference in the exposure in water versus sediments rather than a difference in sensitivity of sediment or water-dwelling species to microplastics. Here, we hypothesize that the relationship between $EEF_{sed,\ CMF}$ and $EEF_{w,\ CMF}$ for cellulosic microfibers would be similar and that a difference would also be due to exposure. Therefore, we use the ratio $EEF_{sed,\ pol}/EEF_{w,\ pol}$ of polymers to derive $EEF_{sed,\ CMF}$ for CMFs from $EEF_{w,\ CMF}$ (see online supplementary material Section 1.1.4 of supporting information A). As sediment-specific effect data become available in the future, $EEF_{sed,\ CMF}$ can be calculated using experimental data.

Characterization factors

Characterization factors combine the fate, exposure, and effect of pollutants to link their emission to their potential impacts on ecosystem quality as an area of protection (Rosenbaum et al., 2007). Characterization factors were calculated at the midpoint (PAF m^3 $d/kg_{emitted}$) level and converted to endpoint following the methodology outlined by Saadi et al. (2025). Conversion of PAF to potentially disappeared fraction of species (PDF) was performed using a severity factor of 1 PDF/PAF (Oginah et al., 2025). In fact, the work of Oginah et al. (2025) showed that this severity factor is applicable when the effect factor is based on an HC20 concentration of EC10 data that lead to population level decline of the tested species (i.e., effects on reproduction, growth, survival), which is the case in our study. Different endpoint indicators are calculated for CFs compatible with multiple impact assessment methods: IMPACT World + (PDF m^2yr/kg), ReCiPe2016 (species yr/kg), and GLAM (PDF yr/kg; Bulle et al., 2019; Huijbregts et al., 2017; UNEP, 2019). The Python code provided in Saadi et al. (2025) was used for the calculations of CFs and their uncertainty.

Case study: cotton versus polyester tee shirts

To demonstrate the relevance of including microfiber impacts in LCA, a case study was conducted comparing two types of tee shirts: one made of 100% cotton and the other of 100% polyester. The objective was to compare the contribution to EQ from the physical effects of leaked microfibers with those from traditional LCIA impact categories such as climate change.

Product description and system boundaries

The functional unit is defined as "wearing a T-shirt 45 times in Québec," in line with the PEFCR Apparel and Footwear guidance (European Commision, 2025). Both tee shirts are assumed to have equal lifespans and undergo the same number of washes (45). A cradle-to-grave approach was applied, covering raw material production, textile manufacturing (fiber extrusion or ginning, spinning, knitting, assembling), use phase (washing and drying), and end-of-life. Transportation between all life cycle stages was included. Detailed system boundaries are provided in online supplementary material Section 1.2 of supporting information A.

Life cycle inventory and microplastic emissions

The model was built in OpenLCA (Ciroth, 2007), using ecoinvent 3.10 and literature data for inventory inputs (Saadi et al., 2025; Sandin et al., 2019; Van Der Velden et al., 2014). Microfiber emissions were estimated for four stages: fiber production, textile manufacturing, use phase (washing), and end-of-life fragmentation. Microfibers can be emitted during the wear of textiles (De Falco et al., 2020); however, these emissions were neglected, as no reliable quantitative data are currently available to represent this process in LCA. Emission rates of textiles and microfibers were based on data from the Plastic Footprint Network (PFN), following the methodology outlined in the textile module of the PFN (Earth Action, 2023). End-of-life fragmentation was calculated using rates from Pinlova and Nowack (2023) while considering exports of textile waste to a low-income country (LIC). A sensitivity analysis was performed by varying the microfiber release rates during the use phase and the mismanagement rates of textiles at the end-of-life. These parameters were applied for two locations, Canada and an average LIC, to assess their influence on the resulting EQ damage. Detailed explanations can be found in online supplementary material supporting information A and calculations of the MP leakage can be found in online supplementary material supporting information B.

LCA software and impact assessment method

The LCA modeling was conducted using OpenLCA v2.4 (Ciroth, 2007), with impact assessment based on IMPACT World + v2.1 (Expert and Midpoint) due to its comprehensive impact categories coverage (Agez et al., 2024; Bulle et al., 2019). All ecosystem quality-related categories were considered, such as climate change and terrestrial ecotoxicity. Assessment of contribution to the climate change impact category was done using OpenLCA results for each life cycle stage. A sensitivity analysis to verify the robustness of EQ damage results was done using ReCiPe2016 v1.03 (endpoint, H), thus ensuring that conclusions were not dependent on the choice of LCIA method. Consistency between both LCIA methods was evaluated in terms of the main impact category contributing to EQ damage and the relative contribution of microfibers.

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Results and discussion

Fate factors

Fate factors in marine water and sediments were computed for six different CMFs (cotton, linen, viscose, lyocell, rayon, modal) and compared with the fate of SMFs (Figure 1). The fate factors in the marine environment range from a few days (for small cellulosic particles) to years (for large synthetic particles). Fate factors for different diameters are available in online supplementary material supporting information B.

Most CMFs are high-density materials, meaning they are nonbuoyant and will settle quickly. Additionally, their degradation rate is much higher than most polymers. Thus, CMFs will spend little time in the water column due to their density and will have smaller fate factors in the sediments than synthetic high-density fibers, as they degrade much faster (S. Kim et al., 2022; Nagamine et al., 2022; Zambrano et al., 2019, 2020). The data used to derive specific surface degradation rates (SSDR) are based on laboratory studies, often under accelerated conditions, which may overestimate environmental degradation, potentially leading to smaller fate factors and characterization factors. Further, Skilbeck et al. (2025) highlight that different fiber treatments can affect their degradation rate. Degradation decreases when fibers are dyed, coated, or finished (e.g., water-repellent treatment). To account for the potential effects of dyes, finishes, or coating, we derived SSDRs using the full range of worst and best-case values reported in the literature (see online supplementary material supporting information B). However, more studies are needed to derive SSDRs or degradation extrapolation factors for each treatment. Therefore, CFs specific to the type of treatment per material are not yet available, but the provided CFs confidence interval should cover this variation. Practitioners of LCA could use the upper limit of the CF intervals for more conservative estimates reflecting much slower degradation of treated textiles.

The fast degradation of CMFs may lead to a faster release of additives. This is particularly relevant given the different types of additives used in cellulosic materials, such as dyes, softeners, or flame retardants, which vary between natural (e.g., cotton) and synthetic (e.g., viscose) sources (Darbra et al., 2012). In contrast, additives in synthetic polymers may remain trapped for longer due to slower degradation rates, although leaching is usually limited by the diffusivity of additives within the polymer bulk (Gulizia et al., 2023). Although studies have examined additive leaching from synthetic microplastics (Sait et al., 2021), direct comparisons with natural and synthetic cellulosic fibers remain nonexistent, limiting our understanding of the leaching and environmental fate of additives across fiber types, which should be quantified for inclusion in LCA (Casagrande et al., 2024; Silva et al., 2025; Sørensen et al., 2021).

SSD

Data were found for 12 aquatic species (both freshwater and marine) over seven phyla and three taxonomic groups (see online supplementary material supporting information B). No EC10 values were found as such in the literature; therefore, the SSD curve is based on LOECs (16 values) and NOECs (11) that were converted to EC10eq (Figure 2). In LCIA, no distinction is made in the sensitivity modeling of freshwater and marine species, mainly due to the lack of data (Fantke et al., 2018; Saadi et al., 2025). Further, because the SSD is based on NOECs and LOECs converted to EC10 using Aurisano et al. (2019) extrapolation factors, which rely mainly on data-rich groups (algae/cyanobacteria, crustaceans, and fish), the resulting distribution may be biased toward these groups and may not reflect the sensitivity of less-studied species.

Mytilus spp., *Gammarus duebeni*, and *Artemia franciscana* were found to be the most sensitive species to CMFs in the dataset, with EC10eq values of 0.0047, 0.0053, and 0.0968 mg/L, respectively (L. Kim et al., 2021; Mateos-Cárdenas et al., 2021; Walkinshaw et al., 2023). The results show that the SSD curves and HC20 values for cellulosic

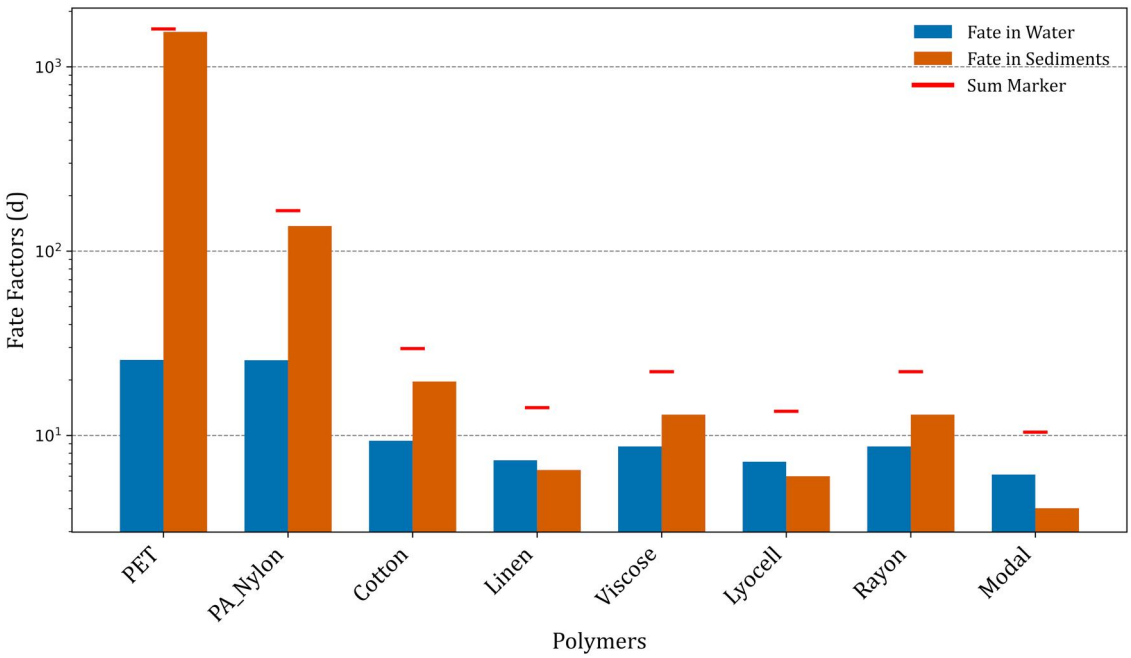

Figure 1 Fate factors (FF) in the marine environment for 10 μm microfibers, for an emission in marine water, shown on a log-scale. The sum represents the total time-integrated mass of microfibers in the marine environment (water and sediments combined). PET = Polyethylene terephthalate; PA = Polyamide.

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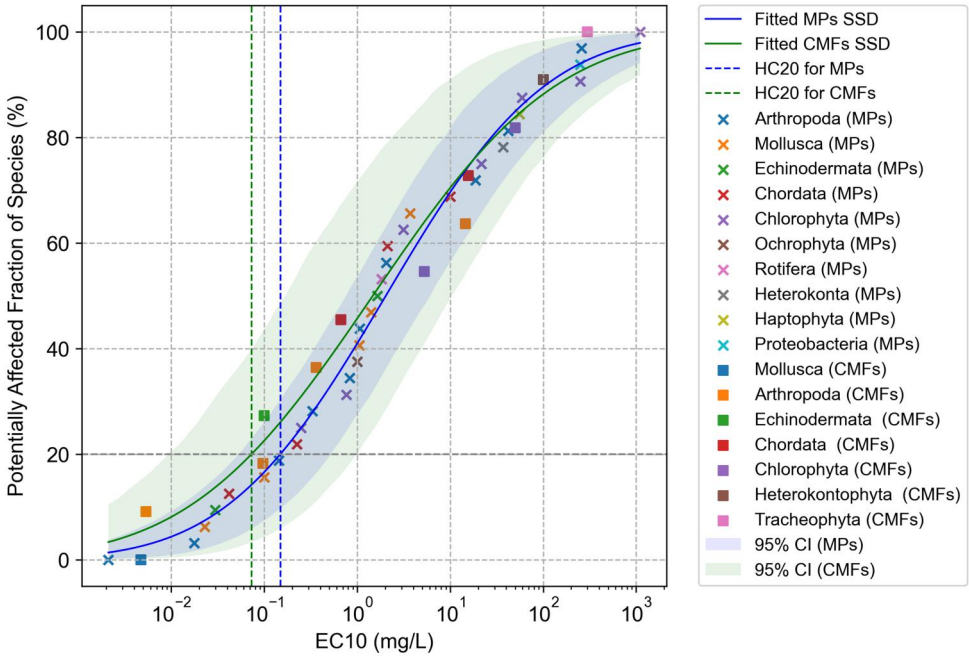

Figure 2 Species sensitivity distribution (SSD) curves for aquatic species exposed to cellulosic microfibers (CMFs, in green) and microplastics (MPs, in blue) with their confidence interval (shaded areas). Each dot represents the average EC10eq (10% effect concentration) for a specific species. The underlying data of the SSD curve of MPs are taken from Corella-Puertas et al. (2023).

microfibers ($HC20_{CMF} = 0.0732$ [0.007 – 0.781] mg/L, Figure 2) and microplastics ($HC20_{MP} = 0.1874$ [0.063 – 0.558] mg/L) are very similar, with no significant difference ($p = 0.58$). The SSD curve for MPs is based on experimental data for virgin MPs that attempted to exclude any MPs containing additives (Corella-Puertas et al., 2023; Lavoie et al., 2022). Further, de Ruijter et al. (2020) have shown that the most suggested and/or proven effect mechanisms for MPs are due to physical effect mechanisms linked to the ingestion of particles (i.e., decreased feeding and internal damage). This could imply that shared physical effects are dominant for both synthetic and cellulosic materials rather than a chemical effect linked to the release of additives. Although CMFs typically contain additives, their toxic contributions may not have manifested in tests on aquatic species due to exposure durations that were too short to allow additive leaching into the water or uptake by tested organisms (Athey et al., 2022; Darbra et al., 2012). Therefore, physical effects likely dominate the species' responses to both particle types in the SSD, at least on a mass basis. Moreover, toxicity needs to be further investigated, as little data are available on the additives added to plastics and anthropogenically modified cellulose, as well as on their fate and effects.

The taxonomic group–split SSD analysis suggests that algae and aquatic plant species are less sensitive (i.e., have higher hazardous concentrations) to MPs and CMFs than vertebrates and invertebrates (Figure 3). This difference may reflect the physical mode of action of MPs and CMFs, which likely depends on ingestion and is less relevant for algae and aquatic plants. A split SSD could not be generated for vertebrates exposed to CMFs, as only two data points were available for this taxonomic group, thus not meeting the minimum required three datapoints for log-normal fitting. The HC20 per taxonomic group data are provided in online supplementary material Table S.3 of supporting information A.

The EEF_{CMF} calculated from the $HC20_{EC10}$ is 2730.9 [256.1 – 29118.0] PAF m^3/kg, larger but in the same order of magnitude and not significantly different from the value for polymers of 1067.5 [358.1–3182.1] PAF m^3/kg (Corella-Puertas et al., 2023). This EEF is applicable to all CMFs, as insufficient data are available to calculate EEFs specific to the size or material. Further, no statistically significant differences were found when enough data were available to compute size, polymer, or shape-specific EEFs for microplastics (Lavoie et al., 2022). Nonetheless, further studies should investigate this once more experimental data are available.

Uncertainty is higher on the EEF of CMFs (two orders of magnitude) than of MPs (one order), due to the more limited number of effect studies available for these materials (Figure 2). As more effect data become available, the robustness of EEF_{CMF} could increase and its value could shift, potentially becoming significantly different from that of MPs, depending on how new data affect the SSD.

Marine characterization factors

Midpoint (PAF_{mar} m^3_{mar} d/kg) and endpoint (PDF m^2yr/kg, Figure 4) marine physical effects on biota CFs for various materials, shapes, and sizes were obtained by combining EEF_{CMF} with FFs (Saadi et al., 2025). Different endpoint indicators (PDF m^2yr/kg, species yr, PDF yr/kg) are calculated for CFs compatible with multiple impact assessment methods. All CFs and their uncertainty values are available in online supplementary material supporting information B. Overall, CFs of textile-sized fibers are one to two orders of magnitude smaller for CMFs than for SMFs, mainly due to the faster degradation of CMFs (Figure 4).

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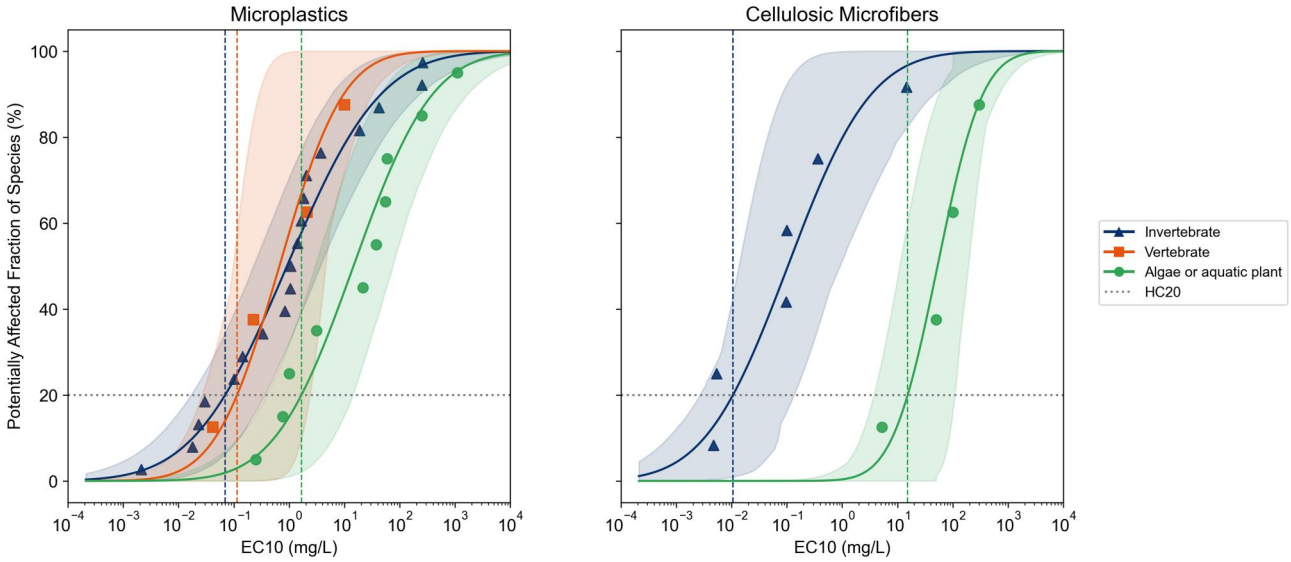

Figure 3. Species sensitivity distribution (SSD) curves split by taxonomic groups (vertebrates, invertebrates, and microalgae or aquatic plants) for microplastics (MPs; left) and cellulosic microfibers (right). The underlying data of the SSD curves of MPs are taken from Corella-Puertas et al. (2023).

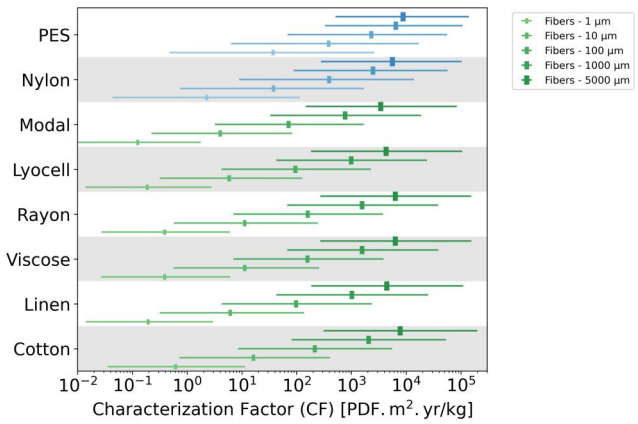

Figure 4. Marine physical effects on biota endpoint characterization factors of microfibers (lines) for an emission in marine water. The microfiber's diameter (1, 10, 100, 1,000, 5,000 μm) is represented by the marker size (e.g., smallest marker = 1 μm microfiber). Large diameters are mostly relevant in the case of regenerated cellulosic fibers. Underlying data can be found in online supplementary information supporting information B. PES = Polyester.

Case studies

The physical effects on biota of emitted microfibers represent between 0.7% (use phase in Québec) and 2.4% (in the average LIC) of the total damage on EQ of the cotton tee shirts when considering all impact categories of IMPACT World+ (Figure 5). In the polyester case, the physical effects represent 6.8% and 21.3% in the Québec and LIC cases, respectively, in line with other LCA studies that have included impacts of SMFs (Pellengahr et al., 2025; Saadi et al., 2025).

Here, we assumed that mismanaged textiles are released into the environment, and we applied a redistribution rate of 5% to estimate the fraction reaching the marine environment before fragmenting (Earth Action, 2023). Impacts in the LIC are higher due to the higher leakage rate and mismanagement rate of textiles (Earth Action, 2023). In high-income countries, a larger portion of MPs accumulates in sewage sludge of wastewater treatment plants, which is often applied to agricultural land, thus transferring MPs to terrestrial ecosystems (Earth Action, 2023; Rolsky et al., 2020). It should be noted that most of the SMF impacts of the polyester tee shirt for a use phase in Canada come from end-of-life fragmentation in an LIC, where textile waste is typically exported. Our CFs exclude terrestrial ecosystems due to the lack of EEF for that compartment, which likely leads to an underestimation of the impacts of leaked fibers, potentially affecting the conclusions of this LCA study. However, the EEF for terrestrial MPs could be applicable to CMFs as well, as it could be hypothesized that the physical effects on terrestrial biota of CMFs and MPs are similar, just as for aquatic biota (Tunali & Nowack, 2025). Terrestrial CFs exist for spherical MPs in terrestrial environments, and the fate model developed could be adapted for synthetic and cellulosic fibers to include them as well (Louvet et al., 2026).

The total damage on EQ of the cotton tee shirt is higher than the damage of the polyester tee shirt, even when including the impacts of leaked fibers. The higher land occupation and transformation, water consumption for agricultural activities, or climate change impacts during production and manufacturing drive this difference. The sensitivity analysis done with ReCiPe2016 shows similar results (see online supplementary material Figure S.4 of supporting information A). The only comparative LCA of synthetic and cellulosic textiles that includes the EQ endpoint also shows higher impacts of cotton (Van Der Velden et al., 2014).

These results suggest that, even in cases with high emissions of CMFs, the ecosystem quality damage of CMFs is much smaller than other life cycle impacts associated with natural fabrics. However, in the synthetic case, the results suggest that the impacts of leaked SMFs are in the same order of magnitude as these other impacts (i.e., climate change, land occupation and transformation, ecotoxicity), but not high enough to change the outcome of the LCA study. These conclusions could also change once terrestrial ecosystems are considered in the CFs of both cellulosic and synthetic fibers. Overall, the results suggest that substituting synthetic with cellulosic textiles does not

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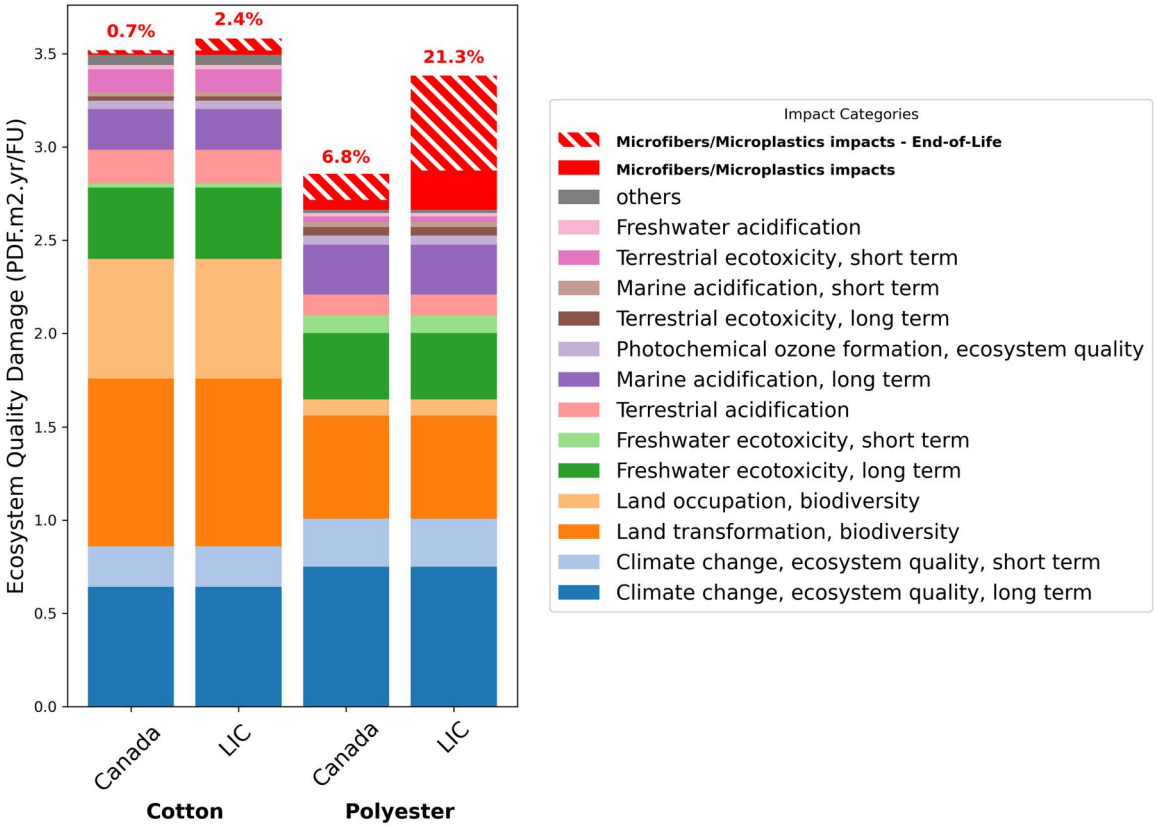

Figure 5. Impacts on ecosystem quality of physical effects on biota of microfibers emitted over the life cycle of a tee shirt in Canada and in a low-income country (LIC), compared to other impact categories. Impacts of microfibers released are shown in red, for a use phase in Canada and in an LIC. Impacts of end-of-life fragmentation of microfibers are shown in dashed red. The percentage in red shows the contribution of microfiber emissions (use and end-of-life) to ecosystem quality damage. PDF = potentially disappeared fraction of species.

reduce life cycle impacts on EQ. However, extending garment lifetimes (i.e., number of wears) could represent a more effective strategy to decrease environmental impacts by reducing the need for new production (Sandin et al., 2019).

Recommendation for future studies on cellulosic fibers

Future research should aim at generating effect data for CMFs specific to soil and sediment environments. So far, only one study has examined the impact of semi-synthetic fibers in soil (Courtene-Jones et al., 2024), and no studies have addressed their effects in sediments. These data would allow for the development of SSD curves and thus EEFs for CMFs specific to these compartments to include them in impact assessment.

Second, the lack of data led to the development of a combined SSD curve for all cellulosic materials (synthetic, natural and regenerated cellulose); however, given their structural and processing differences, natural and regenerated cellulosic fibers could be treated separately in effect modeling. So far, synthetic natural fibers (cotton) and regenerated cellulose materials (lyocell, viscose, processed cellulose fibers) are evenly spread across the SSD curve of CMFs, indicating a similar sensitivity of aquatic species to both types of cellulosic materials. Nonetheless, although both are derived from plant materials, regenerated fibers undergo processing and extrusion into filaments, potentially bringing their physical characteristics closer to synthetic fibers (Athey & Erdle, 2022). Natural fibers, on the other hand, are not extruded and could retain the morphology of their source plants (Fink et al., 2001). This distinction could lead to different physical effects on organisms, potentially resulting in different SSD curves if enough data are available.

Third, further investigation into longer exposure times and chronic endpoints is recommended to determine whether additive-driven toxicity emerges over extended periods, as CMFs degrade and/or additives leach out of them (Athey et al., 2022).

Fourth, animal fibers such as wool, although not cellulosic, represent an important fiber type that requires dedicated physical effect testing. Existing degradation data suggest wool also degrades faster than synthetic fibers (Muniyasamy & Patnaik, 2021), yet it is excluded from microfiber effect experiments. We recommend that future studies develop specific SSD curves for wool for inclusion in LCA.

Finally, more LCA studies that compare synthetic and cellulosic materials should be conducted to put in perspective the physical effects of CMFs and SMFs with other life cycle impacts. In fact, cellulosic materials are often considered more environmentally friendly than plastics, but this is rarely supported by LCA studies. Textile LCA should consider other materials that were not included in our study (cotton and polyester) to assess their environmental impacts while considering the impacts of leaked microfibers. Additional LCA studies

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could also assess other types of products, as microfibers are also released from personal care goods (e.g., cleansing wipes and surgical masks), household carpeting, and cigarette butt filters (Athey et al., 2022; Athey & Erdle, 2022).

Conclusion

This work builds on the methodology of the MarILCA group, which aims to include in LCA the physical effects on biota of microplastics emissions. We developed FFs as well as a EEF_{CMF} for CMFs, which were combined to obtain updated CFs for CMFs, based on an already established LCIA methodology.

The computed FFs show that synthetic fibers are much more persistent than CMFs in the marine environment. Further, the SSD curve for CMFs developed in this work was compared with the curve of MPs, which suggested a similar hazardous concentration and therefore a similar exposure and effect of aquatic organisms to CMFs and MPs, likely due to dominant physical effects. The computed CFs for textile CMFs are one to two orders of magnitude smaller than the CFs of SMFs, mainly due to their much faster degradation in the marine environment. Nonetheless, these results are based on laboratory-generated degradation data, and under real-world conditions, the persistence of cellulosic fibers may be higher, depending also on fiber treatments. The CFs for microfibers were tested in an LCA case study comparing a cotton and a polyester tee shirt, which showed that cotton microfibers emitted over the life cycle of the garment are not a significant contributor to ecosystem quality damage due to their smaller CFs. However, the results show that the impacts of polyester microfibers are a significant contributor in the polyester tee shirt scenario. Further research should investigate the potential impacts of natural and semi-synthetic fibers in soil and sediments, to be included in the LCA of CMFs emitting products and services, as these compartments are not fully considered in impact assessment methods.

This work provides the first SSD comparison of microplastics and cellulosic fibers, and provides the first comparative LCA of a cellulosic and a synthetic textile, while considering microfiber emissions. This study enables future LCAs to account for the impacts of microfibers, supporting a more comprehensive assessment and comparison of cellulosic and synthetic materials that release microfibers throughout their life cycles.

Supplementary material

Supplementary material is available online at *Integrated Environmental Assessment and Management*.

Data availability

Data are available in the supporting information and in the following Zenodo repository: https://doi.org/10.5281/zenodo.15599023.

Author contributions

Nadim Saadi (Conceptualization, Data curation, Formal analysis, Methodology, Software, Visualization, Writing—original draft) and Anne-Marie Boulay (Funding acquisition, Methodology, Supervision, Validation, Writing—review & editing)

Funding

The authors would like to acknowledge the support of the UN Environment Life Cycle Initiative and the Forum for Sustainability through Life Cycle Innovation (FSLCI) to MarILCA and the Canada Research Chairs Program for financial support.

Conflicts of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

References

Acharya, S., Rumi, S. S., Hu, Y., & Abidi, N. (2021). Microfibers from synthetic textiles as a major source of microplastics in the environment: A review. *Textile Research Journal*, *91*, 2136–2156. https://doi.org/10.1177/0040517521991244

Agez, M., Muller, E., Bulle, C., Debarre, L., Seitfudem, G., Viveros Santos, I., & Duval, L. (2024). *IMPACT World+/a globally regionalized method for life cycle impact assessment* [Dataset]. Zenodo. https://zenodo.org/records/14041258

Athey, S. N., Carney Almroth, B., Granek, E. F., Hurst, P., Tissot, A. G., & Weis, J. S. (2022). Unraveling physical and chemical effects of textile microfibers. *Water*, *14*, 3797. https://doi.org/10.3390/w14233797

Athey, S. N., & Erdle, L. M. (2022). Are we underestimating anthropogenic microfiber pollution? A critical review of occurrence, methods, and reporting. *Environmental Toxicology and Chemistry*, *41*, 822–837. https://doi.org/10.1002/etc.5173

Aurisano, N., Albizzati, P. F., Hauschild, M., & Fantke, P. (2019). Extrapolation factors for characterizing freshwater ecotoxicity effects. *Environmental Toxicology and Chemistry*, *38*, 2568–2582. https://doi.org/10.1002/etc.4564

Boucher, J., & Friot, D. (2017). *Primary microplastics in the oceans*. IUCN. https://portals.iucn.org/library/node/46622

Bulle, C., Margni, M., Patouillard, L., Boulay, A.-M., Bourgault, G., De Bruille, V., Cao, V., Hauschild, M., Henderson, A., Humbert, S., Kashef-Haghighi, S., Kounina, A., Laurent, A., Levasseur, A., Liard, G., Rosenbaum, R. K., Roy, P.-O., Shaked, S., Fantke, P., & Jolliet, O. (2019). IMPACT World+: A globally regionalized life cycle impact assessment method. *International Journal of Life Cycle Assessment*, *24*, 1653–1674. https://doi.org/10.1007/s11367-019-01583-0

Casagrande, N., Silva, C. O., Verones, F., Sobral, P., & Martinho, G. (2024). Ecotoxicity effect factors for plastic additives on the aquatic environment: A new approach for life cycle impact assessment. *Environmental Pollution (Barking, Essex: 1987)*, *341*, 122935. https://doi.org/10.1016/j.envpol.2023.122935

Ciroth, A. (2007). ICT for environment in life cycle applications openLCA–A new open source software for life cycle assessment. *International Journal of Life Cycle Assessment*, *12*, 209–210. https://doi.org/10.1065/lca2007.06.337

Corella-Puertas, E., Hajjar, C., Lavoie, J., & Boulay, A.-M. (2023). MarILCA characterization factors for microplastic impacts in life cycle assessment: Physical effects on biota from emissions to aquatic environments. *Journal of Cleaner Production*, *418*, 138197. https://doi.org/10.1016/j.jclepro.2023.138197

Courtene-Jones, W., De Falco, F., Burgevin, F., Handy, R. D., & Thompson, R. C. (2024). Are biobased microfibers less harmful than conventional plastic microfibers: evidence from earthworms. *Environmental Science & Technology*, *58*, 20366–20377. https://doi.org/10.1021/acs.est.4c05856

Downloaded from https://academic.oup.com/ieam/advance-article/doi/10.1093/inteam/vjag007/8419875 by Universite du Quebec a Montreal user on 03 March 2026

Dampe Acharige, T. M., Leusch, F. D. L., & Frid, C. L. J. (2025). Man-made polymers of natural compounds out weight microplastics in Australian seafood: Are we fixating on the wrong thing? *Marine Pollution Bulletin*, *218*, 118160. https://doi.org/10.1016/j.marpolbul.2025.118160

Darbra, R. M., Dan, J. R. G., Casal, J., Àgueda, A., Capri, E., Fait, G., Schuhmacher, M., Nadal, M., Rovira, J., Grundmann, V., Barceló, D., Ginebreda, A., & Guillén, D. (2012). Additives in the textile industry. In B. Bilitewski, R. M. Darbra, & D. Barceló (Eds.), *Global risk-based management of chemical additives I: Production, usage and environmental occurrence* (pp. 83–107). Springer. https://doi.org/10.1007/698_2011_101

De Falco, F., Cocca, M., Avella, M., & Thompson, R. C. (2020). Microfiber release to water, via laundering, and to air, via everyday use: A comparison between polyester clothing with differing textile parameters. *Environmental Science & Technology*, *54*, 3288–3296. https://doi.org/10.1021/acs.est.9b06892

de Ruijter, V. N., Redondo-Hasselerharm, P. E., Gouin, T., & Koelmans, A. A. (2020). Quality criteria for microplastic effect studies in the context of risk assessment: A critical review. *Environmental Science & Technology*, *54*, 11692–11705. https://doi.org/10.1021/acs.est.0c03057

Dos Santos, J. B., Choueri, R. B., Dos Santos, F. E. M., Santos, L. A. D. O., Da Silva, L. F., Nobre, C. R., Cardoso, M. A., De Britto Mari, R., Simões, F. R., Delvalls, T. A., & Gusso-Choueri, P. K. (2024). Are microfibers a threat to marine invertebrates? A sea urchin toxicity assessment. *Toxics*, *12*, 753. https://doi.org/10.3390/toxics12100753

Earth Action. (2023). *Plastic footprint guidelines—Module on microplastic from textile fibres*. http://www.plasticfootprint.earth/

European Commission. (2025). Product Environmental Footprint Category Rules (PEFCR) – Apparel and Footwear version 1.3. https://pefapparelandfootwear.eu/

Fantke, P., Aurisano, N., Bare, J., Backhaus, T., Bulle, C., Chapman, P. M., De Zwart, D., Dwyer, R., Ernstoff, A., Golsteijn, L., Holmquist, H., Jolliet, O., McKone, T. E., Owsianiak, M., Peijnenburg, W., Posthuma, L., Roos, S., Saouter, E., Schowanek, D., … Hauschild, M. (2018). Toward harmonizing ecotoxicity characterization in life cycle impact assessment. *Environmental Toxicology and Chemistry*, *37*, 2955–2971. https://doi.org/10.1002/etc.4261

Fantke, P., Bijster, M., Guignard, C., Hauschild, M. Z., Huijbregts, M. A. J., Jolliet, O., Kounina, A., Magaud, V., Margni, M., Mckone, T., Posthuma, L., Rosenbaum, R. K., van de Meent, D., van Zelm, R., & Fantke, P. (2017). *USEtox 2.0: Documentation (Version 1)*. USEtox International Center. https://doi.org/10.11581/DTU:00000011

Fink, H.-P., Weigel, P., Purz, H. J., & Ganster, J. (2001). Structure formation of regenerated cellulose materials from NMMO-solutions. *Progress in Polymer Science*, *26*, 1473–1524. https://doi.org/10.1016/S0079-6700(01)00025-9

Gulizia, A. M., Patel, K., Philippa, B., Motti, C. A., van Herwerden, L., & Vamvounis, G. (2023). Understanding plasticiser leaching from polystyrene microplastics. *Science of the Total Environment*, *857*, 159099. https://doi.org/10.1016/j.scitotenv.2022.159099

Huijbregts, M. A. J., Steinmann, Z. J. N., Elshout, P. M. F., Stam, G., Verones, F., Vieira, M., Zijp, M., Hollander, A., & van Zelm, R. (2017). ReCiPe2016: A harmonised life cycle impact assessment method at midpoint and endpoint level. *International Journal of Life Cycle Assessment*, *22*, 138–147. https://doi.org/10.1007/s11367-016-1246-y

Kim, D., Kim, H., & An, Y.-J. (2021). Effects of synthetic and natural microfibers on *Daphnia magna*–Are they dependent on microfiber type? *Aquatic Toxicology (Amsterdam, Netherlands)*, *240*, 105968. https://doi.org/10.1016/j.aquatox.2021.105968

Kim, L., Kim, S. A., Kim, T. H., Kim, J., & An, Y.-J. (2021). Synthetic and natural microfibers induce gut damage in the brine shrimp *Artemia franciscana*. *Aquatic Toxicology (Amsterdam, Netherlands)*, *232*, 105748. https://doi.org/10.1016/j.aquatox.2021.105748

Kim, S., Cho, Y., & Park, C. H. (2022). Effect of cotton fabric properties on fiber release and marine biodegradation. *Textile Research Journal*, *92*, 2121–2137. https://doi.org/10.1177/00405175211068781

Lavoie, J., Boulay, A. M., & Bulle, C. (2022). Aquatic micro- and nano-plastics in life cycle assessment: Development of an effect factor for the quantification of their physical impact on biota. *Journal of Industrial Ecology*, *26*, 2123–2135. https://doi.org/10.1111/jiec.13140

Louvet, J., Quik, J. T. K., & Boulay, A.-M. (2026). Regionalized characterization factors for microplastic emissions in life cycle assessment considering multimedia fate modelling. *Journal of Cleaner Production*, *538*, 147217. https://doi.org/10.1016/j.jclepro.2025.147217

Maga, D., Galafton, C., Blömer, J., Thonemann, N., Özdamar, A., & Bertling, J. (2022). Methodology to address potential impacts of plastic emissions in life cycle assessment. *International Journal of Life Cycle Assessment*, *27*, 469–491. https://doi.org/10.1007/s11367-022-02040-1

Mateos-Cárdenas, A., O'Halloran, J., van Pelt, F. N. A. M., & Jansen, M. A. K. (2021). Beyond plastic microbeads—Short-term feeding of cellulose and polyester microfibers to the freshwater amphipod *Gammarus duebeni*. *Science of the Total Environment*, *753*, 141859. https://doi.org/10.1016/j.scitotenv.2020.141859

Muniyasamy, S., & Patnaik, A. (2021). Biodegradable behavior of waste wool and their recycled polyester preforms in aqueous and soil conditions. *Journal of Renewable Materials*, *9*, 1661–1671. https://doi.org/10.32604/jrm.2021.014904

Nagamine, R., Kobayashi, K., Kusumi, R., & Wada, M. (2022). Cellulose fiber biodegradation in natural waters: River water, brackish water, and seawater. *Cellulose*, *29*, 2917–2926. https://doi.org/10.1007/s10570-021-04349-w

Oginah, S. A., Posthuma, L., Hauschild, M., Slootweg, J., Kosnik, M., & Fantke, P. (2023). To split or not to split: Characterizing chemical pollution impacts in aquatic ecosystems with species sensitivity distributions for specific taxonomic groups. *Environmental Science & Technology*, *57*, 14526–14538. https://doi.org/10.1021/acs.est.3c04968

Oginah, S. A., Posthuma, L., Slootweg, J., Hauschild, M., & Fantke, P. (2025). Calibrating predicted mixture toxic pressure to observed biodiversity loss in aquatic ecosystems. *Global Change Biology*, *31*, e70305. https://doi.org/10.1111/gcb.70305

Owsianiak, M., Hauschild, M. Z., Posthuma, L., Saouter, E., Vijver, M. G., Backhaus, T., Douziech, M., Schlekat, T., & Fantke, P. (2023). Ecotoxicity characterization of chemicals: Global recommendations and implementation in USEtox. *Chemosphere*, *310*, 136807. https://doi.org/10.1016/j.chemosphere.2022.136807

Pellengahr, F., Corella-Puertas, E., Mattelin, V., Saadi, N., Bertella, F., Boulay, A.-M., & van der Meer, Y. (2025). Modeling marine microplastic emissions in life cycle assessment: Characterization factors for biodegradable polymers and their application in a textile case study. *Frontiers in Toxicology*, *7*, 1494220. https://doi.org/10.3389/ftox.2025.1494220

Pinlova, B., & Nowack, B. (2023). Characterization of fiber fragments released from polyester textiles during UV weathering. *Environmental Pollution (Barking, Essex: 1987)*, *322*, 121012. https://doi.org/10.1016/J.ENVPOL.2023.121012

Posthuma, L., Sutter, G., & Traas, T. P. (Eds.). (2002). *Species sensitivity distributions in ecotoxicology*. WorldCat.

Quik, J. T. K., Meesters, J. A. J., & Koelmans, A. A. (2023). A multimedia model to estimate the environmental fate of microplastic particles.

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Science of the Total Environment, *882*, 163437. https://doi.org/10.1016/j.scitotenv.2023.163437

Rolsky, C., Kelkar, V., Driver, E., & Halden, R. U. (2020). Municipal sewage sludge as a source of microplastics in the environment. *Current Opinion in Environmental Science & Health*, *14*, 16–22. https://doi.org/10.1016/j.coesh.2019.12.001

Rosenbaum, R. K., Margni, M., & Jolliet, O. (2007). A flexible matrix algebra framework for the multimedia multipathway modeling of emission to impacts. *Environment International*, *33*, 624–634. https://doi.org/10.1016/j.envint.2007.01.004

Royer, S.-J., Greco, F., Kogler, M., & Deheyn, D. D. (2023). Not so biodegradable: Polylactic acid and cellulose/plastic blend textiles lack fast biodegradation in marine waters. *PloS One*, *18*, e0284681. https://doi.org/10.1371/journal.pone.0284681

Saadi, N., Lavoie, J., Fantke, P., Redondo-Hasselerharm, P., & Boulay, A.-M. (2025). Including impacts of microplastics in marine water and sediments in life cycle assessment. *Journal of Cleaner Production*, *520*, 146037. https://doi.org/10.1016/j.jclepro.2025.146037

Sait, S. T. L., Sørensen, L., Kubowicz, S., Vike-Jonas, K., Gonzalez, S. V., Asimakopoulos, A. G., & Booth, A. M. (2021). Microplastic fibres from synthetic textiles: Environmental degradation and additive chemical content. *Environmental Pollution (Barking, Essex: 1987)*, *268*, 115745. https://doi.org/10.1016/j.envpol.2020.115745

Sandin, G., Roos, S., Spak, B., Zamani, B., & Peters, G. (2019). *Environmental assessment of Swedish clothing consumption—six garments, sustainable futures*. https://doi.org/10.13140/RG.2.2.30502.27205

Scientists' Coalition for an Effective Plastics Treaty. (2025). *Policy brief: Article 2 of the draft global plastics treaty text (Definitions)*. IKHAPP. https://ikhapp.org/material/policy-brief-article-2-of-the-draft-global-plastics-treaty-text-definitions/

Siddiqui, S., Hutton, S. J., Dickens, J. M., Pedersen, E. I., Harper, S. L., & Brander, S. M. (2023). Natural and synthetic microfibers alter growth and behavior in early life stages of estuarine organisms. *Frontiers in Marine Science*, *9*. https://www.frontiersin.org/articles/10.3389/fmars.2022.991650

Silva, C. O., Casagrande, N., Candeias, C., Sobral, P., & Martinho, G. (2025). Assessing ecotoxicity of plastic additives to apply in effect factors for Life Cycle Impact Assessment. *Science of the Total Environment*, *967*, 178836. https://doi.org/10.1016/j.scitotenv.2025.178836

Skilbeck, O. J., Blackburn, R. S., & Kay, P. (2025). A review on the biodegradation of textiles in the environment. *Environmental Toxicology and Chemistry*, *9*, vgaf229. https://doi.org/10.1093/etojnl/vgaf229

Sørensen, L., Groven, A. S., Hovsbakken, I. A., Del Puerto, O., Krause, D. F., Sarno, A., & Booth, A. M. (2021). UV degradation of natural and synthetic microfibers causes fragmentation and release of polymer degradation products and chemical additives. *Science of the Total Environment*, *755*, 143170. https://doi.org/10.1016/j.scitotenv.2020.143170

Stanton, T., Johnson, M., Nathanail, P., MacNaughtan, W., & Gomes, R. L. (2019). Freshwater and airborne textile fibre populations are dominated by 'natural', not microplastic, fibres. *Science of the Total Environment*, *666*, 377–389. https://doi.org/10.1016/j.scitotenv.2019.02.278

Traylor, S. D., Granek, E. F., Duncan, M., & Brander, S. M. (2024). From the ocean to our kitchen table: Anthropogenic particles in the edible tissue of U.S. West Coast seafood species. *Frontiers in Toxicology*, *6*, 1469995. https://doi.org/10.3389/ftox.2024.1469995

Tunali, M., & Nowack, B. (2025). Towards including soil ecotoxicity of microplastics and tire wear particles into life cycle assessment. *Ecotoxicology and Environmental Safety*, *303*, 118856. https://doi.org/10.1016/j.ecoenv.2025.118856

UNEP. (2019). *Global Guidance for Life Cycle Impact Assessment Indicators (GLAM) Volume 2*.

Van Der Velden, N. M., Patel, M. K., & Vogtländer, J. G. (2014). LCA benchmarking study on textiles made of cotton, polyester, nylon, acryl, or elastane. *The International Journal of Life Cycle Assessment*, *19*, 331–356. https://doi.org/10.1007/s11367-013-0626-9

Walkinshaw, C., Tolhurst, T. J., Lindeque, P. K., Thompson, R. C., & Cole, M. (2023). Impact of polyester and cotton microfibers on growth and sublethal biomarkers in juvenile mussels. *Microplastics and Nanoplastics*, *3*, 5. https://doi.org/10.1186/s43591-023-00052-8

Watson, K. J., & Wiedemann, S. G. (2019). Review of methodological choices in LCA-based textile and apparel rating tools: Key issues and recommendations relating to assessment of fabrics made from natural fibre types. *Sustainability*, *11*, 3846. https://doi.org/10.3390/su11143846

Weis, J. S., & De Falco, F. (2022). Microfibers: Environmental problems and textile solutions. *Microplastics*, *1*, 626–639. https://doi.org/10.3390/microplastics1040043

Zambrano, M. C., Pawlak, J. J., Daystar, J., Ankeny, M., Cheng, J. J., & Venditti, R. A. (2019). Microfibers generated from the laundering of cotton, rayon and polyester based fabrics and their aquatic biodegradation. *Marine Pollution Bulletin*, *142*, 394–407. https://doi.org/10.1016/j.marpolbul.2019.02.062

Zambrano, M. C., Pawlak, J. J., Daystar, J., Ankeny, M., Goller, C. C., & Venditti, R. A. (2020). Aerobic biodegradation in freshwater and marine environments of textile microfibers generated in clothes laundering: Effects of cellulose and polyester-based microfibers on the microbiome. *Marine Pollution Bulletin*, *151*, 110826. https://doi.org/10.1016/j.marpolbul.2019.110826

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Integrated Environmental Assessment and Management, 2026, 00, 1–10
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